

Reconfigurable Control Scheme for a PV Microinverter Working in Both Grid-Connected and Island Modes

César Trujillo Rodríguez, David Velasco de la Fuente, Gabriel Garcerá, *Member, IEEE*, Emilio Figueres, *Senior Member, IEEE*, and Javier A. Guacaneme Moreno, *Member, IEEE*

Abstract—In this paper, a photovoltaic (PV) microinverter capable of operating in both island mode and grid-connected mode by means of a reconfigurable control scheme is proposed. The main advantage of control reconfiguration is that in grid-connected mode, the microinverter works as a current source in phase with the grid voltage, injecting power to the grid. This is the operation mode of most commercial grid-connected PV microinverters. The idea is to provide those microinverters with the additional functionality of working in island mode without changing their control algorithms for grid-connected mode, which were developed and refined over time. It is proposed that in island mode, the microinverter control is reconfigured to work as a voltage source using droop schemes. These schemes consist in implementing P/Q strategies in the inverters, in order to properly share the power delivered to the loads. The aim of the paper is to show that the proposed control reconfiguration is possible without dangerous transients for the microinverter or the loads. Simulation and experimental results on an 180-W PV microinverter are provided to show the feasibility of the proposed control strategy.

Index Terms—Grid-connected mode, island mode, microinverter, reconfigurable control scheme.

I. INTRODUCTION

IN RECENT years, one of the priorities worldwide is developing alternative sources to produce electric energy, particularly from renewable sources, which produce low environmental contamination levels. Those renewable energy sources play an important role in the long term, and they will give rise to substantial changes in the technologic, environmental, and organizational profile of the global energy system [1].

Manuscript received April 29, 2011; revised September 28, 2011; accepted November 15, 2011. Date of publication November 29, 2011; date of current version November 22, 2012. This work was supported by the Spanish Ministry of Science and Innovation under Grant ENE2009-13998-C02-02.

C. Trujillo Rodríguez and J. A. Guacaneme Moreno are with the Grupo de Sistemas Electrónicos Industriales del Departamento de Ingeniería Electrónica, Universidad Politécnica de Valencia, 46022 Valencia, Spain, and also with the Department of Electronic Engineering, Universidad Distrital Francisco José de Caldas, Bogotá, Colombia (e-mail: cltrujillo@udistrital.edu.co; jguacaneme@udistrital.edu.co).

D. Velasco de la Fuente, G. Garcerá, and E. Figueres are with the Grupo de Sistemas Electrónicos Industriales del Departamento de Ingeniería Electrónica, Universidad Politécnica de Valencia, 46022 Valencia, Spain (e-mail: davede@upv.es; ggarcerá@eln.upv.es; efiguere@eln.upv.es).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TIE.2011.2177615

Moreover, the possibility to produce energy close to the consumption points along with energy delivery problems in isolated and rural areas turn distributed generation (DG) [2] into an interesting and promising technological option.

Accordingly to the expressed above, it is feasible to implement interfaces able to get connected to the grid in order to transfer the energy coming from renewable sources (grid-connected mode), as well as to feed local loads when there is a lack of distribution grid (island mode). Those interfaces are known as microgrids [3].

A microgrid is able to flexibly import and export energy from and to the grid. It controls the flow of active and reactive power [4]. The inverters working in a microgrid must be able to work in grid-connected mode, injecting power to the distribution grid, as well as in island mode, delivering a proper ac voltage quality to local loads [5].

Most commercial grid-connected photovoltaic (PV) microinverters operate as a current source in phase with the grid voltage, injecting power to the grid [6]. The control algorithms for these microinverters have been developed and refined over time, being highly efficient for grid-connected operation. Therefore, many microinverter manufacturing companies may be prone to keep their reliable grid connection control algorithms while incorporating new functions.

The main contribution of this paper is to provide those microinverters with the additional functionality of working in island mode without changing their control algorithms for grid-connected mode.

There are several studies about the proper operation of inverters working in both grid-connected and island modes. A possible solution is based on droop schemes. These schemes consist in implementing P/Q strategies in the inverters, in order to properly share the power delivered to the loads and avoid critical communication lines between inverter modules. In [4] and [7], the inverters are controlled by droop schemes in both operation modes. These inverters are controlled as voltage sources even if they are connected to the grid, so that experienced control algorithms that inject the inverter output current in phase with the grid voltage (current source algorithms) developed for commercial inverters in grid connection mode are discarded.

In [8], it is described that under normal operation, the inverter works in a constant current operation mode in order to provide a preset power to the grid. When the islanding situation

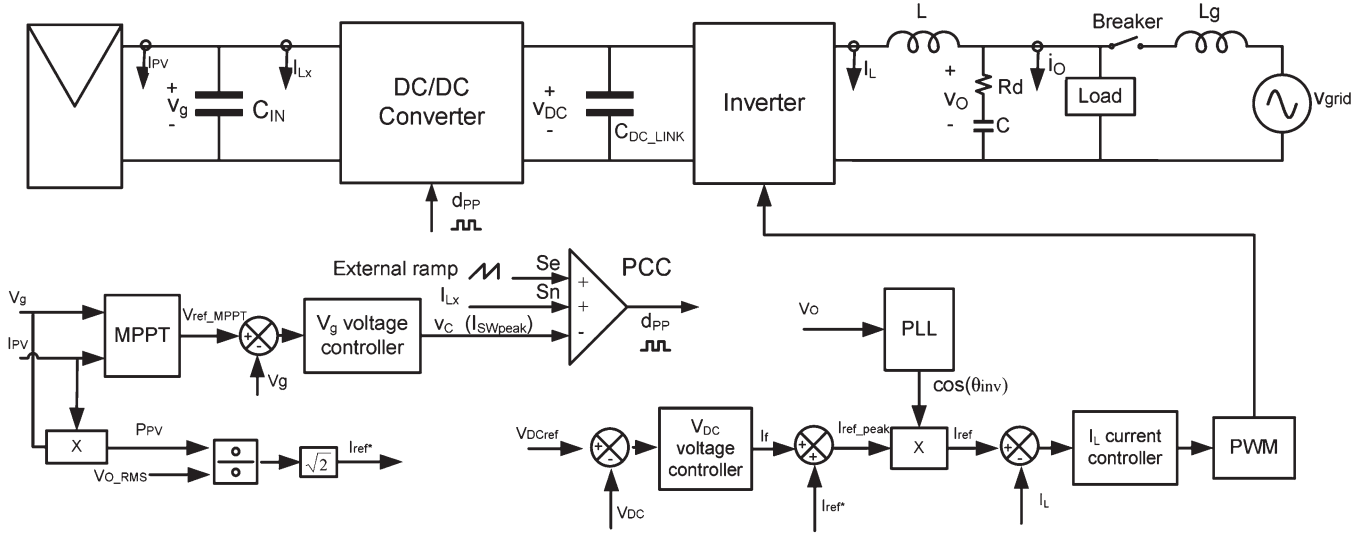


Fig. 1. Block diagram of the PV microinverter. The control configuration corresponds to grid connection mode.

happens, the inverter must detect it and switch to voltage mode operation. In this mode, the islanded inverter will provide a constant voltage to the local load. Additionally, [8] proposes an intelligent load-shedding algorithm for intentional islanding and a synchronization algorithm for grid reconnection. However, the paper does not mention how to parallelize inverters. Furthermore, the reference imposed to the voltage controller has a fixed value.

In [9], it is presented a practical single-phase inverter-based microgrid system that ensures smooth mode transfer between island and grid modes, while maintaining accurate current sharing and high-quality output waveforms. The complete system is implemented using robust CAN communications, including an upper level controller that assigns each inverter to run in voltage or current loop control, a current sharing control, and a mode transfer algorithm. However, this system requires a communications bus to operate properly, which means an increased cost.

In other studies like [10] and [11], reconfigurable control schemes are proposed, based on a very simple and effective type of control. This control is the multiloop linear PI control system. This method uses linear inner and outer PI control loops to regulate the system state variables. However, in these papers, it is not clearly explained how inverters are parallelized sharing the load power.

Given the above, this paper aims to show a reconfigurable control scheme based on multiloop control in both operation modes. Droop schemes are used in the island mode. The idea is to use “current source” control algorithms implemented in commercial microinverters for the grid-connected mode, in which case an inner loop provides control of the ac grid-injected current, whereas an outer loop regulates the dc voltage at the input of the inverter. In island mode, the inner current loop does not change, but the aim of the outer loop is to regulate the inverter output voltage, i.e., a “voltage source algorithm.” The voltage controller follows the reference imposed by droop schemes. These schemes are well suited for parallel inverter operation without communication lines among the inverters sharing the load power.

In addition, this paper presents the operation of the microinverter in island mode, without the need of additional energy storage systems, such as batteries and supercapacitors. The microinverter delivers the energy demanded by the load if enough energy is available from the PV panel.

Additionally, in this paper, it is shown that the proposed control reconfiguration is possible without dangerous transients for the microinverter or the loads.

This paper is organized as follows. The second section presents the characteristics of a PV microinverter working in grid-connected mode. The third section shows the method selected for island detection, which determines whether the microinverter is connected to the distribution grid or not. The fourth section presents the characteristics of the microinverter for island mode operation. The fifth section shows the transition between operation modes. Simulation results are provided in the sixth section to illustrate the feasibility of the proposed controllers. The seventh section shows the experimental results. Finally, conclusions are provided.

II. PV MICROINVERTER SYSTEM DESCRIPTION IN GRID-CONNECTED MODE

Fig. 1 shows the scheme of the PV microinverter that has been implemented. The control configuration corresponds to grid connection mode. The microinverter is fed by a dc programmable source in which the I - V curve of a PV panel has been programmed. The panel voltage is in the range 26–36.7 V at the maximum power point (MPP), so that a dc-dc stage is needed to raise the voltage supplied by the panel to suitable levels ($V_{dc} = 380$ V) for grid connection of the power processing system. A push-pull dc-dc converter has been chosen for this purpose. For dc-ac conversion, a current-controlled H-bridge inverter with bipolar pulse width modulation (PWM) [12] with an output LCL filter [13] has been chosen. Note that the use of a push-pull topology provides galvanic isolation without the need of bulky and heavy low-frequency transformers.

As it can be observed from Fig. 1, the control of the push-pull converter starts from a MPP tracker (MPPT), implemented

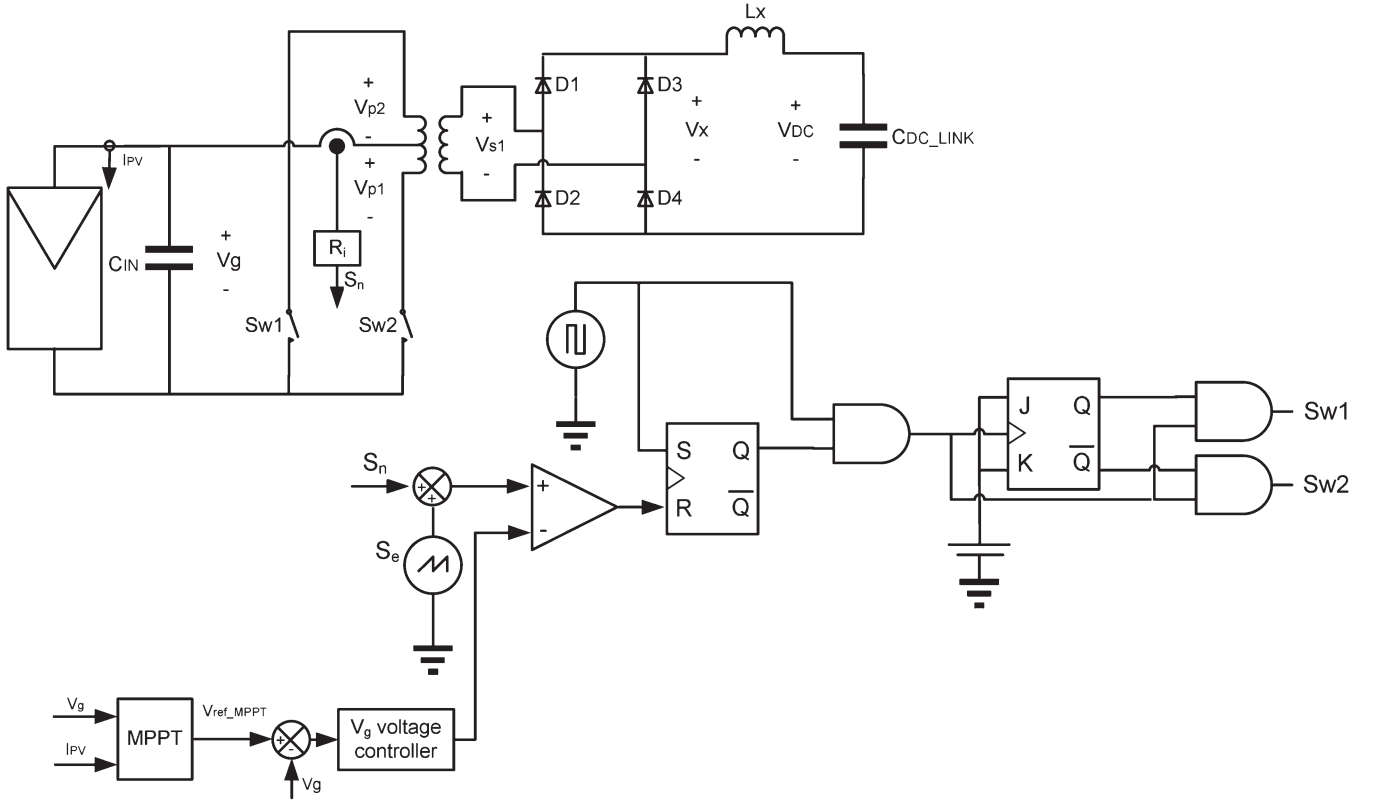


Fig. 2. PCC scheme of the push-pull converter for grid connection mode operation.

by a perturb and observe (P&O) algorithm [14]. The output of the MPPT, V_{g_ref} , is the reference for the PV panel voltage, V_g . The error between both signals is amplified by a PI voltage controller whose output is the control voltage, V_c , of the peak current mode-control loop (PCC) [15]. This control voltage limits the peak value of the current through the primary of the push-pull transformer, avoiding its saturation. The PCC control of the dc-dc converter transformer primary current combined with the control of its input voltage (the PV panel voltage) was proposed in [16] in the context of double-stage PV inverters.

In the PCC PWM modulator, an external stabilization ramp, of slope S_e , is added to the sensed current through the transformer primary (of slope S_n). Fig. 2 shows the control structure of the PCC loop applied to the push-pull converter.

The PWM inverter has an inner current controller based on a harmonic compensator [17] in order to comply with the IEEE 929–2000 standard [18], in terms of the THD of the current injected to the grid. The amplitude, I_{ref_peak} , is the sum the power-feedforward term used to achieve a fast response of the inverter control to changes in the power generated by the panels [19], and the output of the PI controller of the inverter dc-link voltage, I_f . This amplitude is multiplied by $\cos(\theta_{INV})$ for synchronization with the fundamental component of the grid voltage. A synchronous rotating reference frame phase-locked loop (dqPLL) [20] is used for grid synchronization. Fig. 3 shows the dqPLL block diagram.

Table I shows the electrical parameters of the PV microinverter under study.

Table II shows the expression of the chosen controllers for both, the inner current loops and the outer voltage loops,

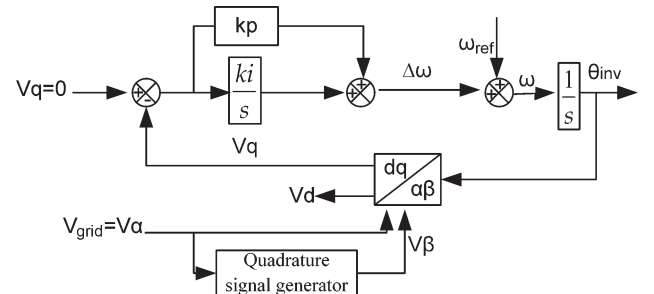


Fig. 3. Block diagram of the dqPLL.

TABLE I
ELECTRICAL PARAMETERS OF THE PV MICROINVERTER UNDER STUDY

Parameter	Values
Panel voltage variation (V_g)	[24V – 36.7V]
Injected power by panel (P_{pv_MPP})	180W
DC-link voltage (V_{DC})	380V
Inverter output voltage (V_{O_RMS})	230V _{RMS}
Inverter output frequency (f_g)	50Hz
Push-pull inductance (L_X)	11mH
Push-pull input capacitance (C_{IN})	470μF
Push-pull transformer turns ratio (N)	20
Push-pull switching frequency (f_s)	20kHz
DC-link capacitance (C_{DC_LINK})	1mF
Inverter inductance (L)	19.1mH
Inverter output capacitance (C)	600nF
Damping resistance (R_d)	50Ω
Inverter switching frequency (f_{st})	20kHz

along with the corresponding crossover frequencies and phase margins both for the push-pull and for the H-bridge inverter

TABLE II
EXPRESSIONS OF THE CHOSEN CONTROLLERS, CROSSOVER FREQUENCIES, AND PHASE MARGINS. GRID-CONNECTED MODE

Push-pull			
Controller	Expression in the continuous time	C. freq.	P. mar.
PWM modulator gain. R_i . S_e .	$F_M = \frac{L_X \cdot f_s \cdot N^2}{(1-2 \cdot D) \cdot V_g \cdot R_i + L_X \cdot S_e \cdot N^2}$ 0.015Ω 2181 V/s	2.63kHz	66.4°
Input voltage controller	$-1 \cdot \left(k_p + \frac{k_i}{s} \right) = -1 \cdot \left(1.5 + \frac{300}{s} \right)$	98.4Hz	89.6°
Inverter			
Current controller	$k_p + \frac{k_h \cdot B_h \cdot s}{s^2 + B_h \cdot s + (\omega_h^2)} = 1.04 + \frac{100 \cdot (2\pi) \cdot s}{s^2 + 2\pi \cdot s + (100\pi)^2}$	1.28kHz	58.6°
DC_link voltage controller	$-1 \cdot \left(k_p + \frac{k_i}{s} \right) = -1 \cdot \left(2.5 + \frac{0.04}{s} \right)$	10Hz	87°

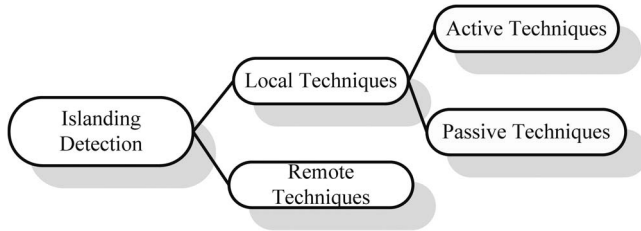


Fig. 4. Anti-islanding techniques classification.

in grid connection operation mode. In Table II, the term F_M is the PWM modulator gain, which depends on S_e and S_n . This value should be adjusted properly in order to guarantee the stability of the current loop [16]. Details about the adjustment of the inverter controllers in grid-connected mode can be found in [21].

III. ISLANDING DETECTION METHODS

The condition of “islanding” in a distributed power generation system is an electrical phenomenon which occurs when the energy supplied by the power grid is interrupted and the distributed generators (DGs) continue energizing some or the entire load. Thus, the power grid stops controlling this isolated part of the distribution system, which contains both loads and generation, so that security, restoration of service, and reliability of the equipment may be compromised [22], [23].

The main idea to detect an islanding situation is to supervise the DG output parameters and/or other system parameters in order to determine if changes indicating an islanding condition have occurred. Islanding detection techniques can be divided into remote and local ones, whereas the latter techniques are divided into passive and active ones, as shown in Fig. 4 [24].

Before defining the different methods of islanding detection, it is important to highlight two key features in order to understand the islanding phenomenon. The first one is associated with the so-called “nondetection zone” (NDZ). NDZ can be defined as the range (in terms of the power difference between DG inverter and load or load parameters) in which an islanding

detection scheme under test fails to detect this condition [25]. The second one is associated with the type of loads (potential loads inside an isle), which can be modeled as a parallel RLC circuit. This circuit is primarily used because it raises more difficulties for islanding detection techniques than others. Generally, nonlinear loads that produce current harmonics, or constant power loads do not represent difficulties for islanding detection [26].

The passive techniques are based on islanding detection through monitoring of parameters such as voltage, current, frequency, and/or their characteristics.

Those techniques interrupt the inverter operation when a transition occurs beyond the limits established for these parameters. They have the advantage of not worsening the quality of power, but exhibit a considerable NDZ. The main passive techniques are

- over/undervoltage and over/underfrequency [27];
- detection of voltage and current harmonics [28];
- detection based on state estimators [29].

The active techniques intentionally introduce disturbances at the output of the inverter to determine if they affect voltage, frequency and impedance parameters, in which case, it is assumed that the grid has been disconnected and the inverter is isolated from the load.

Active techniques have the advantage of remarkably reducing or even eliminating the NDZ, but in order to achieve their purpose, they may deteriorate the quality of the grid voltage or even cause instability. Among the active techniques the following can be found:

- Harmonic injection/detection of impedance [30].
- Sandia frequency shift (SFS) Sandia voltage shift (SVS) [31].
- Variation of active power and reactive power [32].
- General electric frequency schemes (GEFS) [33].

The method used in this work is the active method based on harmonic injection. This method has the particularity of monitoring changes in the grid impedance due to the injection of a particular harmonic or a subharmonic [34]. It is based on

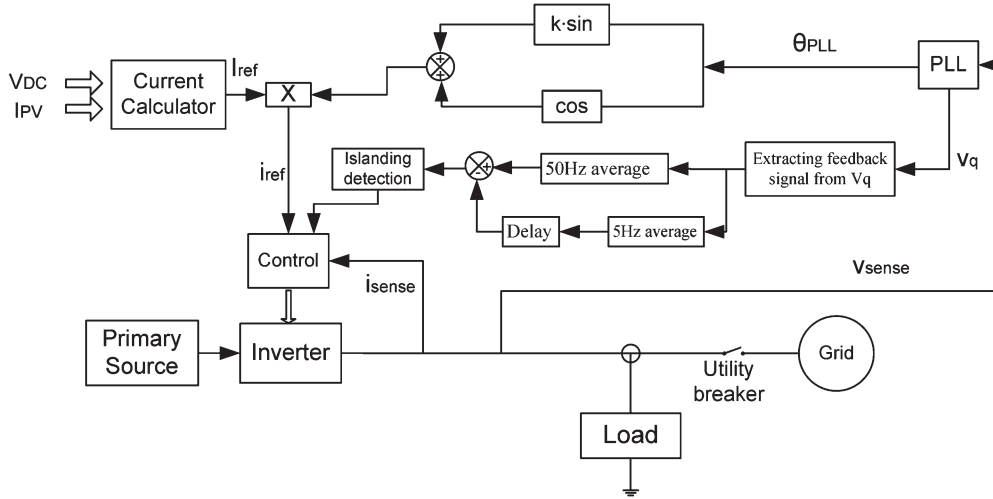


Fig. 5. Block diagram of the adopted islanding detection method.

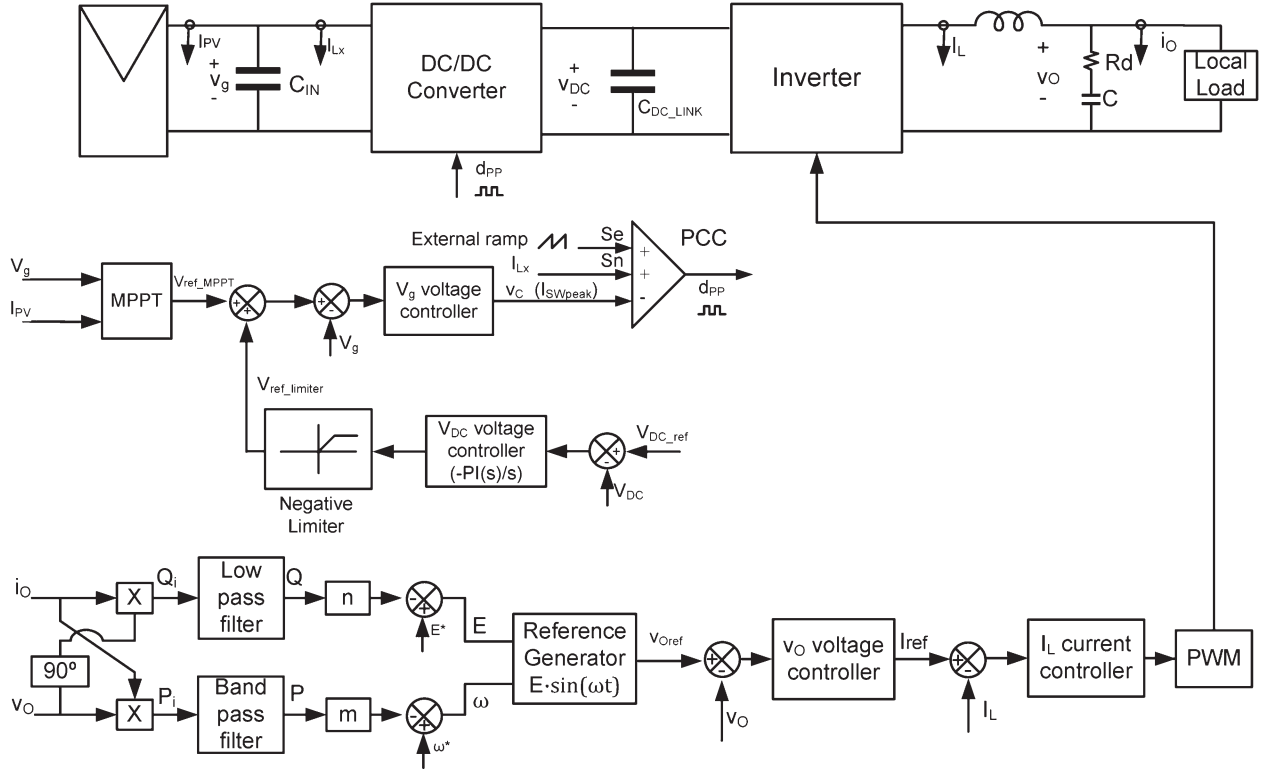


Fig. 6. Control configuration of the PV microinverter working in island mode.

the idea that when grid is connected, the impedance of the grid at the injected harmonic frequency is lower than that of the load, so that the current produced by this harmonic flows into the grid. When the grid is disconnected, the current produced by the injected harmonic flows through the load, producing a voltage harmonic which can be detected.

In order to inject a second harmonic, the method based on PLL perturbation shown in [30] has been adopted in this work. The idea is to inject a harmonic current, in this case a second-order harmonic, to change the phase of the inverter current ($\cos(\theta_{INV})$ in Fig. 1). This is achieved by injecting a sinusoidal signal synchronized with the PLL phase which forces to smoothly modify the phase of the inverter current. A

feedback signal is extracted from the point common coupling voltage (V_q component) as a result of the injected harmonic signal. When the grid is disconnected, a high second harmonic in V_q can be detected.

Fig. 5 shows a block diagram of the adopted anti-islanding method based on the injection of a second harmonic and on the detection of grid impedance changes.

IV. MICROINVERTER SYSTEM DESCRIPTION IN ISLAND MODE

The control configuration proposed for the PV microinverter working in island mode is shown in Fig. 6.

TABLE III
EXPRESSIONS OF THE CHOSEN CONTROLLERS, CROSSOVER FREQUENCIES, AND PHASE MARGINS. ISLAND MODE

Push-pull			
Controller	Expression in the continuous time	C. freq.	P. mar.
PWM modulator gain R_i	$F_M = \frac{L_X \cdot f_s \cdot N^2}{(1-2 \cdot D) \cdot V_g \cdot R_i + L_X \cdot S_e \cdot N^2}$	1.42kHz	77.7°
S_e	0.015Ω 2181 V/s		
Push-pull input voltage controller	$-1 \cdot \left(k_p + \frac{k_i}{s} \right) = -1 \cdot \left(3.75 + \frac{750}{s} \right)$	199Hz	90.2°
DC_link voltage controller	$-\frac{1}{s} \cdot \left(K_p + \frac{K_i}{s} \right) = -\frac{1}{s} \cdot \left(30299 + \frac{5756.81}{s} \right)$	50Hz	76.2°
Inverter			
Current controller	$k_p + \frac{k_h \cdot B_h \cdot s}{s^2 + B_h \cdot s + (\omega_h^2)} = 1.04 + \frac{100 \cdot (2\pi) \cdot s}{s^2 + 2\pi \cdot s + (100\pi)^2}$	1.6kHz	81.2°
Output voltage controller	$K_p + \frac{K_i}{s + \delta} = 0.048 + \frac{1384}{s + 10}$	502Hz	65.2°

On the contrary, in island mode, the inverter feeds the local loads, offering similar waveform characteristics, both in voltage and frequency, as the distribution grid. As the inverter controls its output voltage in island mode, it is obvious that it cannot also regulate its input dc voltage. Apparently, the dc-dc converter could achieve this task in a conventional way, i.e., by closing a voltage control loop of its output voltage (i.e., the inverter input voltage) around the inner current loop. However, in that case the PV panel voltage would become uncontrolled, and the extracted PV energy would not be efficiently managed. Thus, the MPP or another power point demanded by the loads would not be reached. With the approach proposed in [35], the push-pull converter can regulate simultaneously both its output and input voltages. The input voltage is adjusted by closing a control loop around the inner current loop, whereas the output voltage is regulated by means of an additional control loop, as it is shown in Fig. 6. It is worth pointing out that the controller of V_{dc} in Fig. 6 is a PI one with a negative sign in cascade with an integrator.

Moreover, it should be taken into account that the power provided by the microinverter is that required by the loads. For this reason, it is necessary to find the inverter voltage reference as a function of the active and reactive power consumed by the load. The droop method [36] is used to meet this goal.

Table III shows the expressions of the chosen regulators for both the inner current loops and the outer voltage loops, along with the corresponding crossover frequencies and phase margins for the microinverter in island mode. Details about the adjustment of the controllers in island mode can be found in [35], [37].

In order to establish the inverter voltage reference for the voltage controller, droop schemes [38] are applied. Such methods are based on the well-known concept of the power flow management in large-scale ac generators, which lies in reducing the frequency when the output power is increased [39].

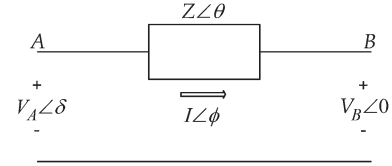


Fig. 7. Power flow through a transmission line.

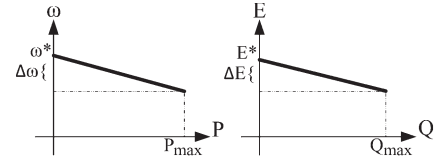


Fig. 8. Static droop characteristics for inductive output impedance.

A scheme that represents the power flow through a transmission line is shown in Fig. 7.

Where Z and θ represent the magnitude and phase of the output impedance, respectively, and δ represents the phase of the voltage between points A and B .

When the impedance of the transmission line has an inductive behavior ($Z = jX$), the active power (P) and reactive power (Q) supplied to the load can be expressed as

$$P = \frac{V_A \cdot V_B}{X} \sin \delta \quad (1)$$

$$Q = \frac{V_A \cdot V_B \cdot \cos \delta - V_B^2}{X} \quad (2)$$

From (1) and (2), and considering small a phase difference between V_A and V_B (i.e., a small δ), it can be observed that P has a strong dependence on the angle δ , whereas Q depends on the amplitude difference between V_A y V_B .

The adopted droop scheme is shown in Fig. 8.

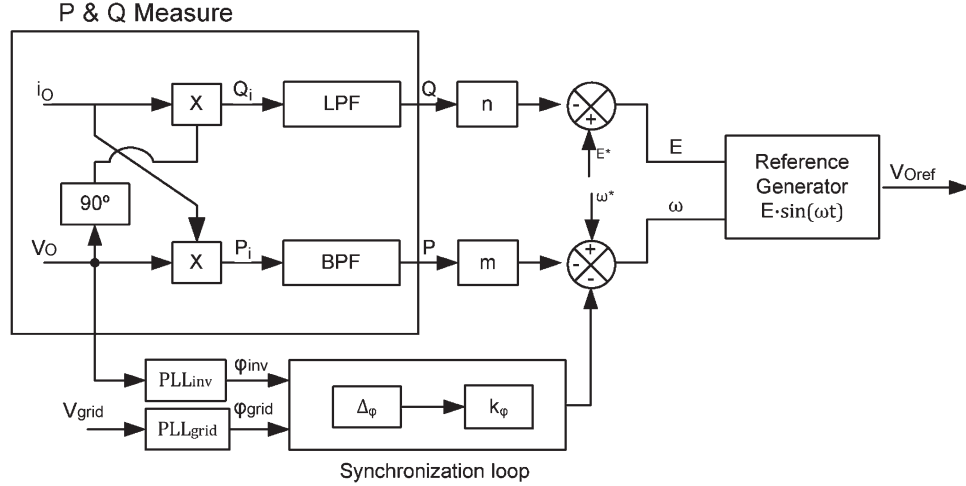


Fig. 9. Block diagram of the droop scheme with synchronization loop.

The frequency (ω) and the voltage amplitude (E) of the inverter voltage reference can be expressed through (3) and (4)

$$\omega = \omega^* - m \cdot P \quad (3)$$

$$E = E^* - n \cdot Q \quad (4)$$

Being ω_* and E^* the frequency and amplitude of the inverter reference voltage at not load, whereas m and n represent the frequency and amplitude droop coefficients, respectively.

The coefficients m and n can be chosen as in the conventional droop method to ensure steady-state control objectives [40] as follows:

$$m = \frac{\Delta\omega}{P_{\max}} \quad (5)$$

$$n = \frac{\Delta E}{Q_{\max}} \quad (6)$$

where $P_{\max}$ and $Q_{\max}$ are the maximum active and reactive powers that can be delivered by the inverter, and $\Delta\omega$ and ΔE are the maximum frequency and output voltage amplitude deviations allowed. Hence, these two parameters can be fixed by the designer taking into account the tradeoff between the power sharing accuracy and the frequency and amplitude deviation. In this case, m and n were set to 0.001.

The active power (P) can be obtained by means of low-pass filtering the inverter instantaneous output power. It is necessary take into account that the conventional droop scheme cannot satisfy the need for synchronization with other inverters or with the grid due to the frequency variation of the inverters, which produce a phase deviation. To obtain the required performance, it is necessary to eliminate the dc component of the active power. In [41] is proposed include a high-pass filter in series with a low-pass filter (LPF). Therefore, the resulting filter is a band pass filter (BPF).

The reactive power (Q) is calculated in a similar way. However, in this case, the output voltage measurement is delayed 90° , and a LPF is used. The expressions of the chosen BPF(s) and of the LPF(s) are presented in (7) and (8), respectively

$$BPF(s) = \frac{s}{0.3183 \cdot s + 1} \cdot \frac{1}{31.83 \cdot 10^{-3} \cdot s + 1} \quad (7)$$

$$LPF(s) = \frac{1}{31.83 \cdot 10^{-3} \cdot s + 1} \quad (8)$$

V. TRANSITION BETWEEN OPERATION MODES

In this section, it is provided a brief description of the control configuration changes that occur in the transition between operation modes.

A. Transition From Grid-Connected Mode to Island Mode

In the transition from the grid-connected mode to island mode, it is necessary to change the voltage controller of the inverter, because in grid-connected mode, the voltage loop controls the dc_link voltage, V_{dc} , whereas in island mode, the load voltage, V_o , is controlled. Furthermore, the reference for the voltage controller in island mode is provided by the droop scheme, whereas it is a constant value for the grid-connected mode. In the same way, the push-pull voltage controller changes its control configuration to regulate the dc_link voltage and adjust the input voltage to the push-pull.

In order to avoid abrupt transitions from one operation mode to the other, the final conditions of the controllers in a particular mode must be equal to the initial conditions of the controller in the new mode. It should be highlighted that the same output current controller for both push-pull and inverter is used in both operation modes [42].

B. Transition From Island Mode to Grid-Connected Mode

When the microinverter is operating in island mode and the grid is reconnected, it is possible to interconnect the microinverter to the grid. This is done by synchronizing the microinverter output phase with the grid phase. This action is accomplished by measuring the corresponding phases and doing their difference near to zero using a synchronization loop [41]. Fig. 9 shows the synchronization loop.

From Fig. 9, it can be found the expression of the inverter output frequency during the synchronization with the grid

$$\omega = \omega^* - m \cdot (P \cdot BPF(s)) - k_\phi \cdot \Delta_\phi \quad (9)$$

where Δ_ϕ is the phase difference (in radians) between the microinverter and the grid, and k_ϕ is the frequency proportional constant. For this work, the phase measurement was performed

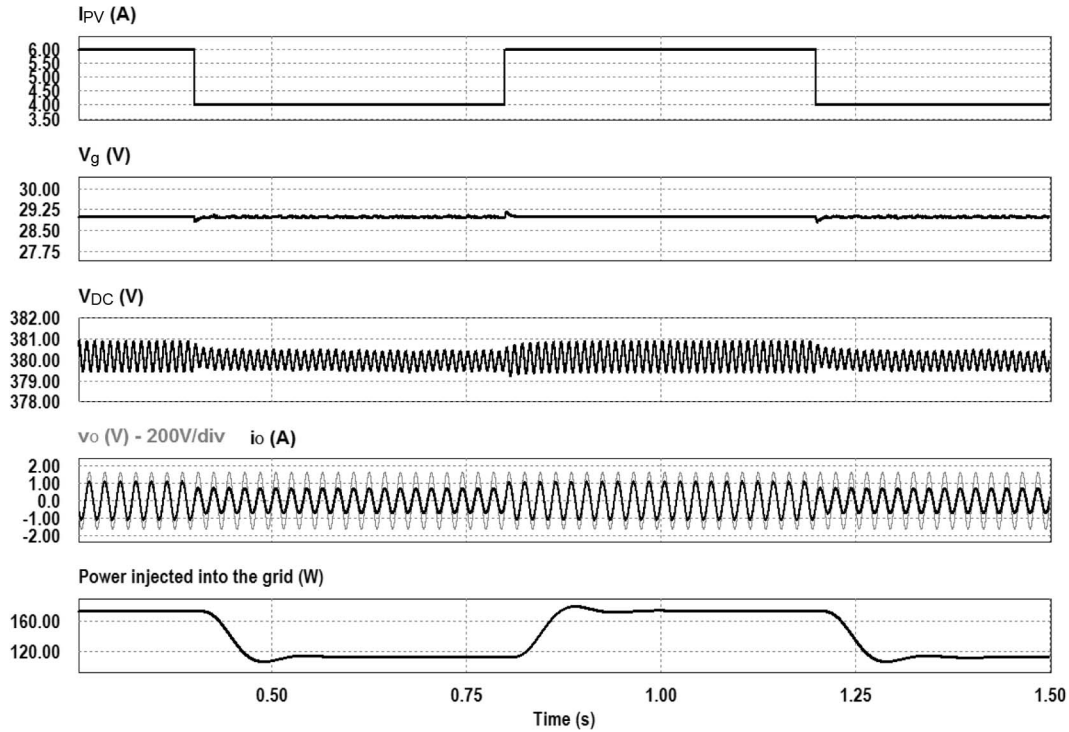


Fig. 10. Main waveforms in grid-connected mode.

by means of two dqPLL. Once the angular difference becomes zero through the action of the synchronization loop, the switch reconnects the microinverter to the grid. A value of $k_\varphi \approx 0.2$ was chosen to ensure a fast and suitable synchronization between the inverter and the grid without affecting the system stability.

VI. SIMULATION RESULTS

The results of the PV microinverter operating in grid-connected mode and in island mode are shown in the following. The system has been simulated by means of PSIM 7.0.5 simulation software [43]. Additionally, it has been considered a grid inductance (L_g) of 1.91 mH, i.e., 10% of the inverter output inductor and 0.2% of the base inductance of a 180 W PV inverter.

Fig. 10 shows the main waveforms of the microinverter connected to the grid in the case of step variations of the current delivered by the PV source, I_{pv} . It is worth pointing out that irradiance variations on the PV panels cannot provoke so quick I_{pv} variations in a real case, but this simulation is a tough stability test for the microinverter control loops. In the experimental results to be shown in Section VII, those I_{pv} steps can be achieved by means of a dc programmable source. In Fig. 10, it is observed that under variations of the PV source current, the output current is properly controlled, allowing the power injection to the grid according to the maximum available power.

Fig. 11 shows the waveforms when the grid is disconnected from the microinverter and from a 440 Ω resistive load and the waveforms after the subsequent microinverter control reconfig-

uration from grid-connected mode to island mode. The upper waveform signals the islanding instant, going from high to low level when the grid is disconnected.

As it can be observed from Fig. 11, the microinverter injects the maximum available power before the disconnection of the grid (180 W). When the grid is disconnected, the control scheme changes to island mode, feeding the microinverter a resistive load that absorbs 120 W at 230 Vrms. It can be observed that the PV panel voltage, V_g , increases in order to leave the panel MPP.

Fig. 12 shows the waveforms during the reconnection of the microinverter to the grid, and the subsequent microinverter control reconfiguration from island mode to grid-connected mode.

Note that the reconnection transient exhibits no oscillations, and the controllers are adjusted quickly to operate in grid connection mode.

Fig. 13 shows the main waveforms of two equal microinverters operating in parallel when the disconnection between the grid and the microinverters takes place. In this situation, the microinverters have a load distribution defined by their corresponding m and n slopes, which agree in this case. The upper waveform signals the islanding event.

Fig. 14 shows the main waveforms of two equal microinverters operating in parallel when the reconnection between the microinverters and the grid takes place. In $t = 0.4$ sec., the grid voltage comes back, and the synchronization algorithm starts to synchronize v_{O1} , v_{O2} , and v_{grid} . In $t = 1$ sec., the grid circuit breaker is closed.

The transition from island mode to grid mode is very smooth, without oscillations that can damage in the system.

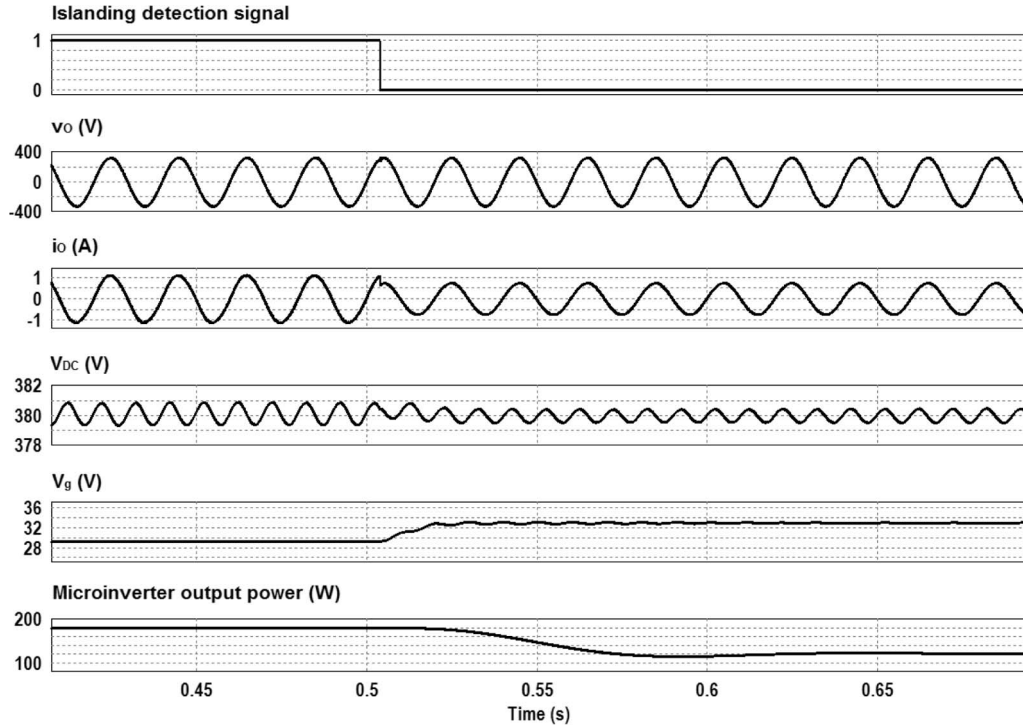


Fig. 11. Main waveforms during the microinverter control reconfiguration from grid-connected mode to island mode.

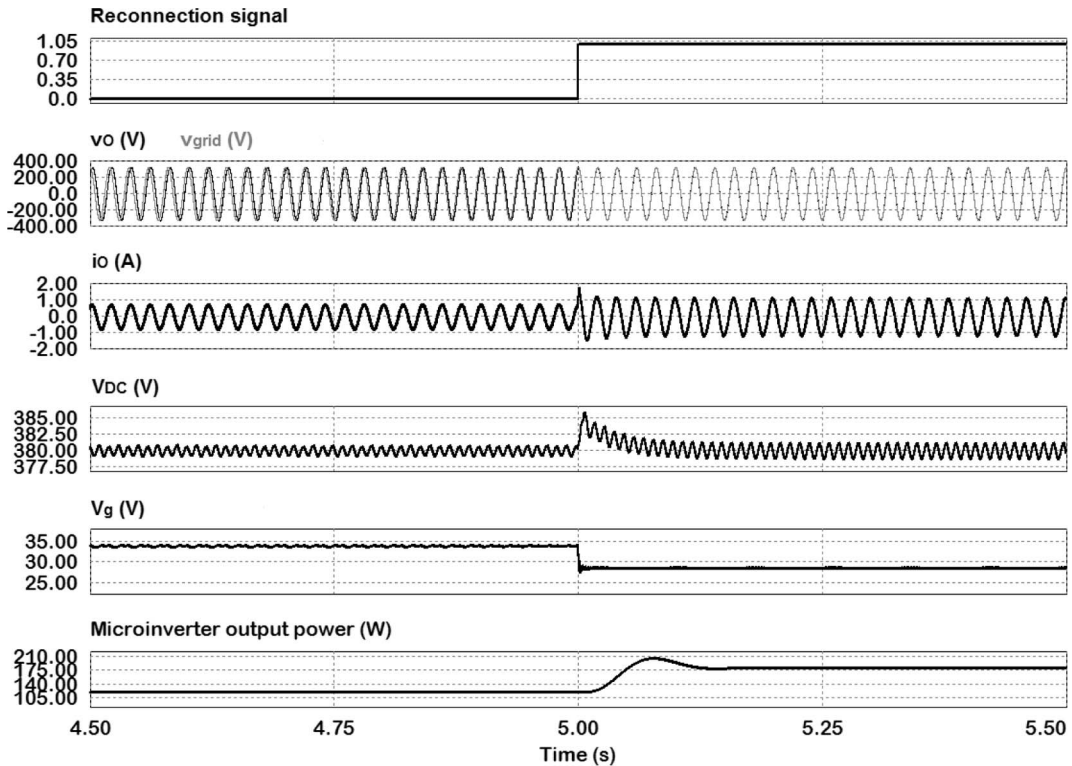


Fig. 12. Main waveforms during the microinverter control reconfiguration from island mode to grid-connected mode.

VII. EXPERIMENTAL RESULTS

Experimental tests have been performed in order to show the feasibility of the proposed controllers. The hardware setup shown in Fig. 15 is based on an 180-W microinverter whose characteristics have been summarized in Table I, whereas the adjustment of the controllers has been described in previous

sections. A TMS320F28335 Digital Signal Processor (DSP) performs the inner current loop and an outer PI controller that ensures voltage regulation for both grid mode and island mode, corresponding to the inverter control. The droop schemes of the inverter and the voltage controller of the push-pull converter are also running on the DSP. The sampling frequency

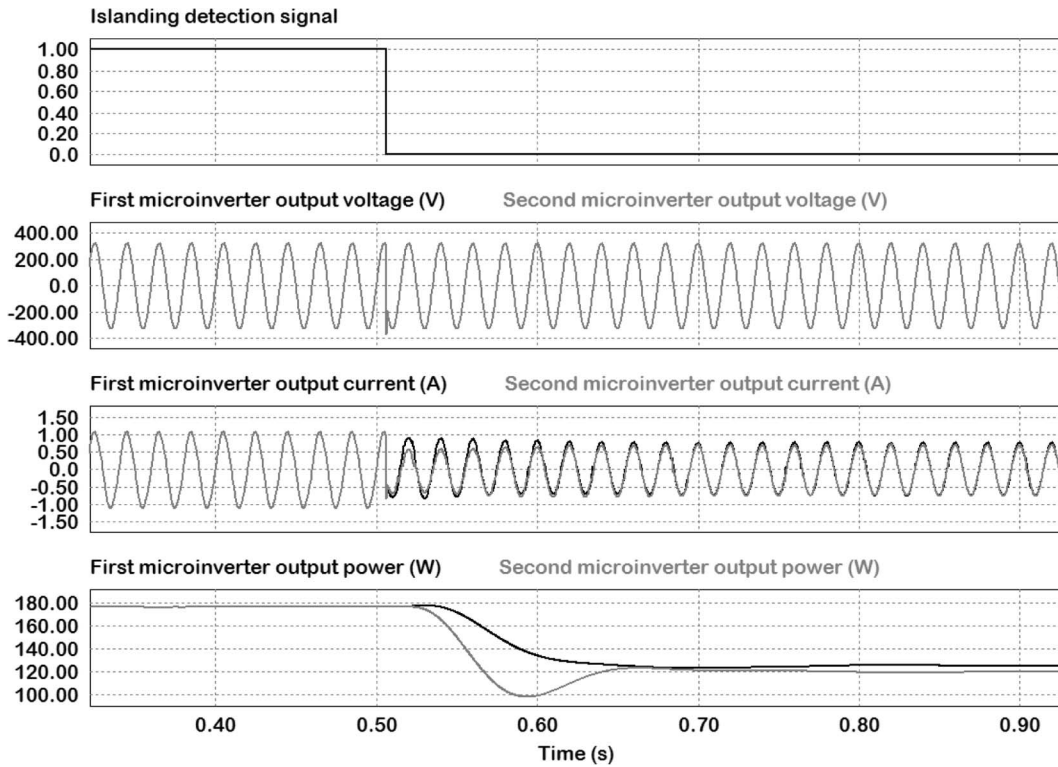


Fig. 13. Main waveforms of two inverters connected in parallel in the transition from the grid-connected mode to the island mode.

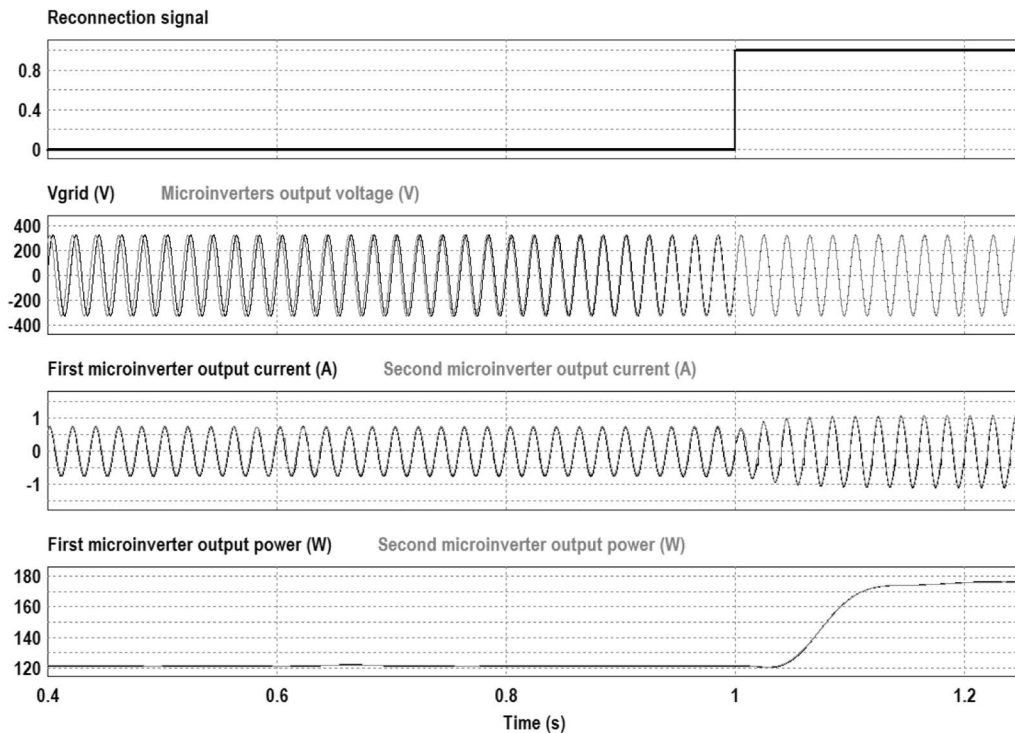


Fig. 14. Main waveforms of two inverters connected in parallel in the transition from the island mode to the grid-connected mode.

is 40 kHz, and the Tustin method has been chosen for the controller discretization. The DSP control also includes a PLL block in order to synchronize the microinverter with the grid. The experimental results have been obtained by connecting a programmable dc power supply AMREL SPS800-12-D-013 to the microinverter input.

Fig. 16 shows a large-signal stability test of the microinverter in grid-connected mode. The dc power supply has been configured as a current source that undergoes a sudden step in I_{PV} from 4 A to 6 A and back to 4 A. The output voltage of the dc supply (microinverter input voltage) is controlled by the push-pull to a constant value $V_{dc} = 28$ V. The test is useful

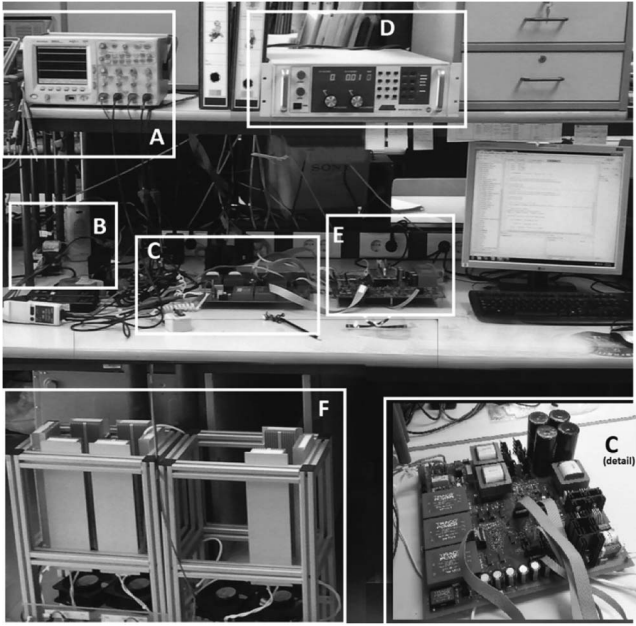


Fig. 15. Laboratory setup to test the reconfigurable control scheme for a microinverter working in both grid-connected and island modes. (a) Oscilloscope. (b) Point of common coupling. (c) Microinverter. (d) dc power supply AMREL SPS800-12-D-013, E. TMS320F28335 Digital Signal Processor. (f) Loads.

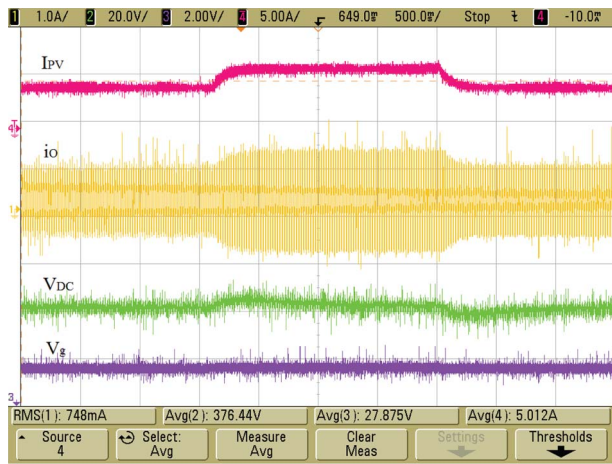


Fig. 16. Large-signal response of the microinverter. Input current step I_{PV} (first trace, 5A/div) from 4A to 6A and back to 4A. i_O (second trace, 1A/div). V_{dc} (third trace, 20V/div). V_g (fourth trace, 2V/div). Time scale = 500 ms/div.

to demonstrate the robustness of the dynamics of the proposed control structure to fast transients, although the variations of I_{PV} are not so fast in a real PV panel. It can be observed from Fig. 16 that the dynamic response of both the grid-injected current and of V_{dc} is stable and well damped.

Fig. 17 shows the transition from grid-connected mode to island mode (start of the islanding situation) with a local load of $350\ \Omega$ (151 W at 230 Vrms). As it can be observed, the microinverter can, with a smooth transient, switch from grid-connected mode to island mode while regulating the output voltage quickly to the nominal operating range. In grid-connected mode, the microinverter output current follows a reference that allows to inject the maximum available power from the programmable dc source, whereas in island mode,

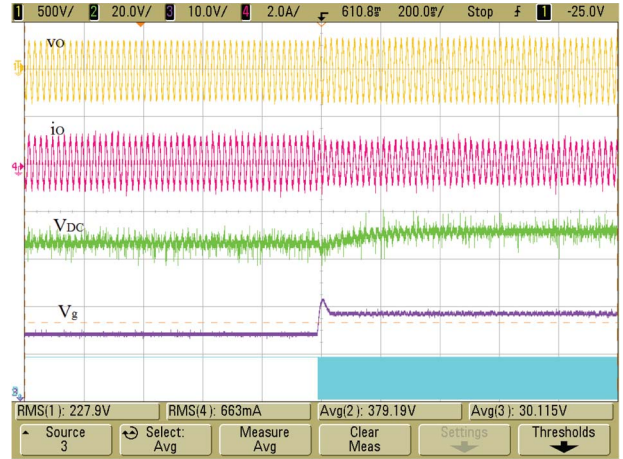


Fig. 17. Transient response of the transition from grid-connected mode to island mode. v_O (first trace, 500 V/div). i_O (second trace, 2 A/div). V_{dc} (third trace, 20 V/div). V_g (fourth trace, 10 V/div). Islanding detection signal (fifth trace). Time scale = 200 ms/div.

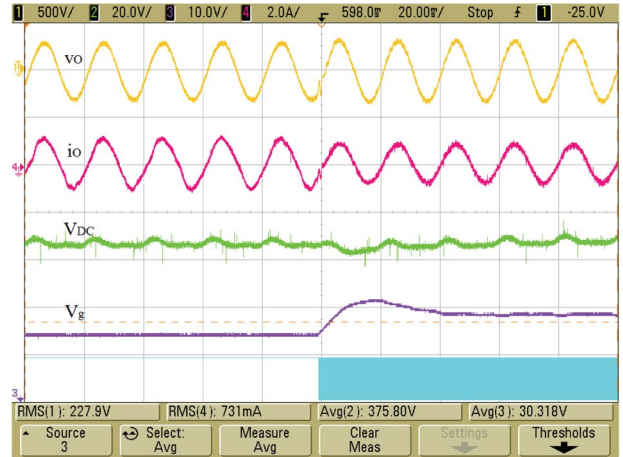


Fig. 18. Detail of the transient response of the transition from grid-connected mode to island mode. v_O (first trace, 500 V/div). i_O (second trace, 2 A/div). V_{dc} (third trace, 20 V/div). V_g (fourth trace, 10 V/div). Islanding signal (fifth trace). Time scale = 20 ms/div.

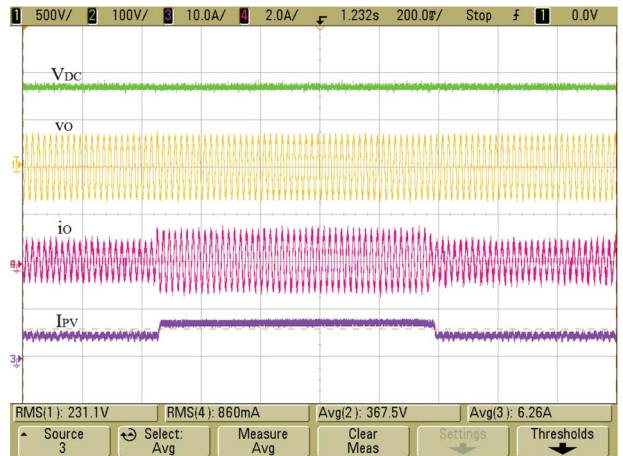


Fig. 19. Transient response for load step changes in island mode. V_{dc} (first trace, 100 V/div). v_O (second trace, 500 V/div). i_O (third trace, 2 A/div). I_{PV} (fourth trace, 10 A/div). Time scale = 200 ms/div.

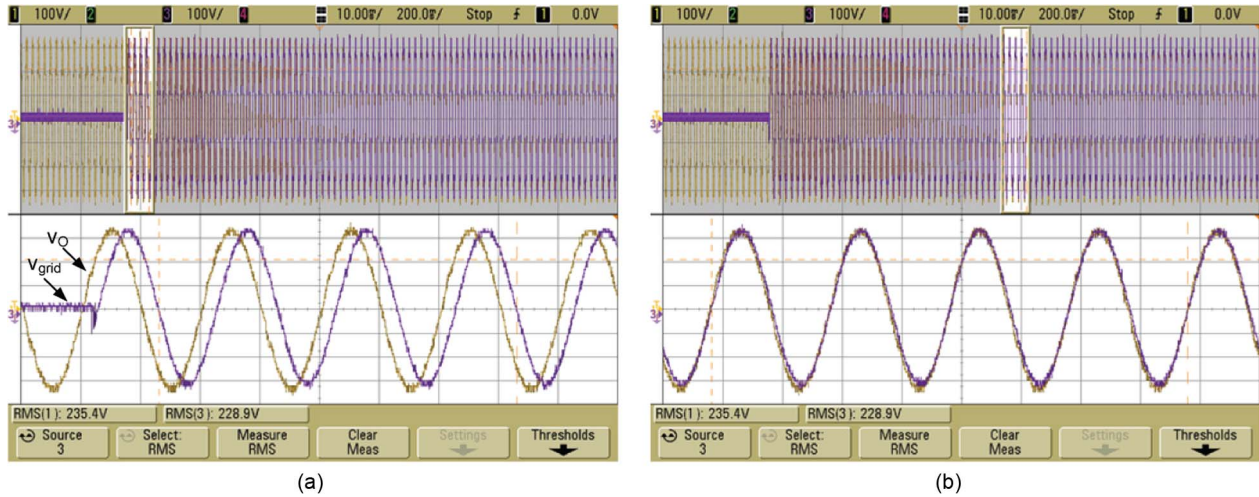


Fig. 20. Microinverter output voltage synchronization with the grid voltage (top, 100 V/div). Detail of the synchronization signals. (a) Start of the synchronization. (b) Synchronized signals. Time scale = 200 ms/div.

the output current follows the reference imposed by the droop schemes (load). In this case, the resistive load absorbs a smaller power (151 W) than that injected to the grid before islanding (180 W). Additionally, the microinverter input voltage is adjusted by a P&O MPPT algorithm [14] to follow the load power consumption.

Fig. 18 shows a detail of Fig. 17, in which a good transient response between both operation modes is observed.

It is observed from Fig. 17 that the steady-state value of V_{dc} in island mode is slightly higher (385 V) than in grid-connected mode (380 V). This small difference is due to slight differences in the V_{dc} reference imposed to the inverter and to the dc-dc converter in the experimental prototype.

Fig. 19 shows the transient response of the microinverter for resistive load step changes from 120 W to 182 W and back to 120 W (440 Ω to 290 Ω and back to 440 Ω) in island mode. It is shown that when the microinverter works with the droop schemes, the load changes do not affect significantly the microinverter operation, whereas the transients are smooth.

Fig. 20 shows the synchronization between the microinverter output voltage and the grid voltage when the grid is reconnected. When both phases match, the circuit breaker can be reclosed.

Fig. 21 shows a smooth transition between island mode and grid-connected mode. Once the grid and the inverter voltages are synchronized, the grid circuit breaker is reclosed. This transition has no significant voltage or current transients. The microinverter is loaded with 150 W in island mode. After the transition, the microinverter injects the power supplied by the programmable dc source into the grid, in this case 180 W.

Fig. 22 shows a detail of Fig. 21, in which a good transient response of the transition from island mode to grid-connected mode, is observed.

VIII. CONCLUSION

The control algorithms for grid-connected PV microinverters have been developed and refined over the time, being highly reliable for grid-connected operation. The main contribution of

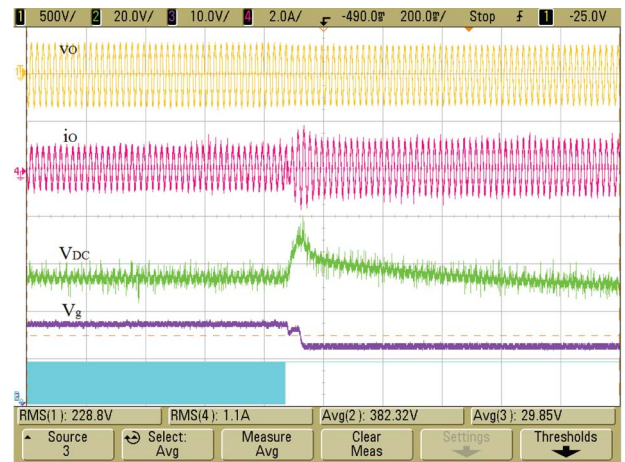


Fig. 21. Transient response for the transition from island mode to grid-connected mode. v_O (first trace, 500 V/div). i_O (second trace, 2 A/div). V_{dc} (third trace, 20 V/div). V_g (fourth trace, 10 V/div). Grid reconnection signal (fifth trace). Time scale = 200 ms/div.

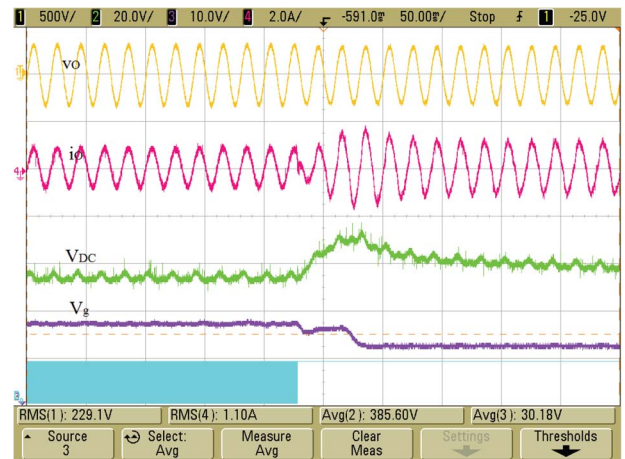


Fig. 22. Detail of the transient response of the transition from island mode to grid-connected mode. v_O (first trace, 500 V/div). i_O (second trace, 2 A/div). V_{dc} (third trace, 20 V/div). V_g (fourth trace, 10 V/div). Grid reconnection signal (fifth trace). Time scale = 50 ms/div.

this paper is to provide those microinverters with the additional functionality of working in island mode without changing their control algorithms for grid-connected mode. It is proposed to reconfigure the microinverter control in island mode to work as a voltage source using droop schemes. This paper is focused on PV generation systems based on a double power conversion structure (dc-dc converter + inverter), although the reconfiguration proposal could be applied to other topologies.

The control structures for both PV microinverter operation modes are presented in this paper. In the grid-connected mode, the inverter controls its input dc voltage and its output current, working as a current source in phase with the grid voltage, injecting the available PV power to the grid, whereas the dc-dc converter connected between the PV panel and the inverter controls the microinverter input voltage (i.e., the PV panel voltage) achieving the MPP tracking. This is the operation mode of most commercial PV grid-connected microinverters based on a double power conversion structure. In the island mode, the same current controller is used for the inverter, but its voltage controller is reconfigured in order to regulate the inverter output voltage. Moreover, in that operation mode, the dc-dc converter controls its input voltage, whereas its output voltage (i.e., the inverter input voltage) is regulated by means of an additional control loop. In that way, proper voltage and frequency levels are provided to the local loads. The reference for the inverter voltage loop is obtained by means of droop schemes. Such schemes allow the inverter to work in grid isolated mode. The reconfiguration of the controllers from one mode to the other one is validated through simulation. Finally, experimental results have been presented in order to validate the proposed approach, showing a proper reconfiguration of the controllers both when the islanding situation appears and when the grid is reconnected.

REFERENCES

- [1] "Renewable 2009, Global Status Report," Worldwatch Institute Washington, DC, 2009.
- [2] H. B. Puttgen, P. R. MacGregor, and F. C. Lambert, "Distributed generation: Semantic hype or the dawn of a new era?" *IEEE Power Energy Mag.*, vol. 1, no. 1, pp. 22–29, Jan./Feb. 2003.
- [3] H. Kanchev, L. Di, F. Colas, V. Lazarov, and B. Francois, "Energy management and operational planning of a microgrid with a PV-based active generator for smart grid applications," *IEEE Trans. Ind. Electron.*, vol. 58, no. 10, pp. 4583–4592, Oct. 2011.
- [4] J. M. Guerrero, J. C. Vasquez, J. Matas, M. Castilla, and L. G. de Vicuna, "Control strategy for flexible microgrid based on parallel line-interactive UPS systems," *IEEE Trans. Ind. Electron.*, vol. 56, no. 3, pp. 726–736, Mar. 2009.
- [5] J. C. Vasquez, J. M. Guerrero, A. Luna, P. Rodriguez, and R. Teodorescu, "Adaptive droop control applied to voltage-source inverters operating in grid-connected and islanded modes," *IEEE Trans. Ind. Electron.*, vol. 56, no. 10, pp. 4088–4096, Oct. 2009.
- [6] Y.-M. Chen, H.-C. Wu, Y.-C. Chen, K.-Y. Lee, and S.-S. Shyu, "The ac line current regulation strategy for the grid-connected PV system," *IEEE Trans. Power Electron.*, vol. 25, no. 1, pp. 209–218, Jan. 2010.
- [7] K. Jaehong, J. M. Guerrero, P. Rodriguez, R. Teodorescu, and N. Kwanghee, "Mode adaptive droop control with virtual output impedances for an inverter-based flexible ac microgrid," *IEEE Trans. Power Electron.*, vol. 26, no. 3, pp. 689–701, Mar. 2011.
- [8] I. J. Balaguer, L. Qin, Y. Shuitao, U. Supatti, and F. Z. Peng, "Control for grid-connected and intentional islanding operations of distributed power generation," *IEEE Trans. Ind. Electron.*, vol. 58, no. 1, pp. 147–157, Jan. 2011.
- [9] C. Chien-Liang, W. Yubin, J.-S. Lai, Y.-S. Lee, and D. Martin, "Design of parallel inverters for smooth mode transfer microgrid applications," *IEEE Trans. Power Electron.*, vol. 25, no. 1, pp. 6–15, Jan. 2010.
- [10] K. Hyosung, Y. Taesik, and C. Sewan, "Indirect current control algorithm for utility interactive inverters in distributed generation systems," *IEEE Trans. Power Electron.*, vol. 23, no. 3, pp. 1342–1347, May 2008.
- [11] M. A. Hassan and M. A. Abido, "Optimal design of microgrids in autonomous and grid-connected modes using particle swarm optimization," *IEEE Trans. Power Electron.*, vol. 26, no. 3, pp. 755–769, Mar. 2011.
- [12] N. Mohan, T. Undeland, and W. Robbins, *Power Electronics: Converters, Applications, and Design*. Hoboken, NJ: Wiley, 1995.
- [13] G. Shen, X. Zhu, J. Zhang, and D. Xu, "A new feedback method for PR current control of LCL-filter-based grid-connected inverter," *IEEE Trans. Ind. Electron.*, vol. 57, no. 6, pp. 2033–2041, Jun. 2010.
- [14] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, "A technique for improving P&O MPPT performances of double-stage grid-connected photovoltaic systems," *IEEE Trans. Ind. Electron.*, vol. 56, no. 11, pp. 4473–4482, Nov. 2009.
- [15] D. M. Sable and R. B. Ridley, "Comparison of performance of single-loop and current-injection control for PWM converters that operate in both continuous and discontinuous modes of operation," *IEEE Trans. Power Electron.*, vol. 7, no. 1, pp. 136–142, Jan. 1992.
- [16] G. Garcera, R. González-Medina, E. Figueres, and J. Sandia, "Dynamic modeling of DC-DC converters with peak current control in double-stage photovoltaic grid-connected inverters," *Int. J. Circuit Theory Appl.*, Jan. 2011.
- [17] M. P. Kazmierkowski, R. Krishnan, and F. Blaabjerg, *Control in Power Electronics*. London, U.K.: Academic, 2002.
- [18] *IEEE Recommended Practice for Grid Interface of Photovoltaic (PV) Systems*, IEEE Std. 929, 2000.
- [19] M. Ciobotaru, R. Teodorescu, and F. Blaabjerg, "Control of single-stage single-phase PV inverter," in *Proc. IEEE Eur. Conf. Power Electron. Appl.*, 2005, pp. 1–10.
- [20] M. Ciobotaru, R. Teodorescu, and F. Blaabjerg, "A new single phase PLL structure based on second order generalized integrator," in *Rec. IEEE PES*, Jeju, Korea, 2006, pp. 1511–1516.
- [21] N. A. Ninad and L. A. C. Lopes, "Operation of single-phase grid-connected inverters with large dc bus voltage ripple," in *Proc. IEEE EPC*, Oct. 25–26, 2007, pp. 172–176.
- [22] A. Pigazo, M. Liserre, R. A. Mastromauro, V. M. Moreno, and A. Dell'Aquila, "Wavelet-based islanding detection in grid-connected PV systems," *IEEE Trans. Ind. Electron.*, vol. 56, no. 11, pp. 4445–4455, Nov. 2009.
- [23] L. Soo-Hyoung and P. Jung-Wook, "New islanding detection method for inverter-based distributed generation considering its switching frequency," *IEEE Trans. Ind. Appl.*, vol. 46, no. 5, pp. 2089–2098, Sep./Oct. 2010.
- [24] P. Mahat, Z. Chen, and B. Bak-Jensen, "Review of islanding detection methods for distributed generation," in *Proc. 3rd Int. Conf. Elect. Utility Deregulation Restructuring Power Technol.*, China, Apr. 6–9, 2008, pp. 2743–2748.
- [25] F. De Mango, M. Liserre, A. D. Aquila, and A. Pigazo, "Overview of anti-islanding algorithms for PV systems. Part I: Passive methods," in *Proc. 12th Int. Conf. EPE-PEMC*, Portoroz, Slovenia, 2006, pp. 1878–1883.
- [26] V. Task, "Evaluation of islanding detection methods for photovoltaic utility interactive power systems," Int. Energy Agency, Paris, France, Tech. Rep. IEAPVPS T5-09/2002, Mar. 2002.
- [27] H. H. Zeineldin and J. L. Kirtley, "A simple technique for islanding detection with negligible nondetection zone," *IEEE Trans. Power Del.*, vol. 24, no. 2, pp. 779–786, Apr. 2009.
- [28] S. Jang and K. Kim, "An islanding detection method for distributed generations using voltage unbalance and total harmonic distortion of current," *IEEE Trans. Power Del.*, vol. 19, no. 2, pp. 745–752, Apr. 2004.
- [29] M. Liserre, A. Pigazo, A. Dell'Aquila, and V. M. Moreno, "An anti-islanding method for single-phase inverters based on a grid voltage sensorless control," *IEEE Trans. Ind. Electron.*, vol. 53, no. 5, pp. 1418–1426, Oct. 2006.
- [30] M. Ciobotaru, V. G. Agelidis, R. Teodorescu, and F. Blaabjerg, "Accurate and less-disturbing active antiislanding method based on PLL for grid-connected converters," *IEEE Trans. Power Electron.*, vol. 25, no. 6, pp. 1576–1584, Jun. 2010.
- [31] X. Wang, W. Freitas, W. Xu, and V. Dinavahi, "Impact of DG interface controls on the sandia frequency shift antiislanding method," *IEEE Trans. Energy Convers.*, vol. 22, no. 3, pp. 792–794, Sep. 2007.

- [32] C. Jeraputra and P. N. Enjeti, "Development of a robust anti-islanding algorithm for grid interconnection of distributed fuel cell powered generation," *IEEE Trans. Power Electron.*, vol. 19, no. 5, pp. 1163–1170, Sep. 2004.
- [33] S. Huili, L. A. C. Lopes, and L. Zhixiang, "Analysis and comparison of islanding detection methods using a new load parameter space," in *Proc. 30th IEEE/IECON*, Korea, 2004, vol. 2, pp. 1172–1177.
- [34] A. V. Timbus, R. Teodorescu, F. Blaabjerg, and U. Borup, "Online grid measurement and ENS detection for PV inverter running on highly inductive grid," *IEEE Power Electron. Lett.*, vol. 2, no. 3, pp. 77–82, Sep. 2004.
- [35] C. L. Trujillo, D. Velasco, E. Figueres, G. Garcerá, and R. Ortega, "Modeling and control of a push-pull converter for photovoltaic microinverters operating in island mode," *Appl. Energy*, vol. 88, no. 8, pp. 2824–2834, Aug. 2011.
- [36] J. M. Guerrero, L. Hang, and J. Uceda, "Control of distributed uninterruptible power supply systems," *IEEE Trans. Ind. Electron.*, vol. 55, no. 8, pp. 2845–2859, Aug. 2008.
- [37] R. Ortega, G. Garcerá, E. Figueres, O. Carranza, and C. L. Trujillo, "Design and application of a two degrees of freedom control with a repetitive controller in a single phase inverter," in *Proc. IEEE ISIE*, Jun. 27–30, 2011, pp. 1441–1446.
- [38] J. M. Guerrero, J. Matas, L. G. de Vicuna, M. Castilla, and J. Miret, "Wireless-control strategy for parallel operation of distributed-generation inverters," *IEEE Trans. Ind. Electron.*, vol. 53, no. 5, pp. 1461–1470, Oct. 2006.
- [39] W. Yao, M. Chen, J. Matas, J. M. Guerrero, and Q. Zhao-Ming, "Design and analysis of the droop control method for parallel inverters considering the impact of the complex impedance on the power sharing," *IEEE Trans. Ind. Electron.*, vol. 58, no. 2, pp. 576–588, Feb. 2011.
- [40] A. Tuladhar, H. Jin, T. Unger, and K. Mauch, "Parallel operation of single phase inverter modules with no control interconnections," in *Proc. IEEE-APEC Conf.*, 1997, pp. 94–100.
- [41] J. M. Guerrero, L. G. de Vicuna, J. Matas, J. Miret, and M. Castilla, "A wireless controller for parallel inverters in distributed online UPS systems," in *Proc. 29th IEEE/IECON*, Nov. 2003, vol. 2, pp. 1637–1642.
- [42] T. Thacker, F. Wang, and D. Boroyevich, "Islanding control of a distributed generation unit's power conversion system to the electric utility grid," in *Proc. PESC*, Jun. 2005, pp. 210–216.
- [43] *PSIM 7.0.5 User's Guide*, Powersim Inc., Woburn, MA, Mar. 2006.



Gabriel Garcerá (S'97–A'99–M'04) received the Ingeniero de Telecomunicación (M.Sc.) and Dr. Ingeniero de Telecomunicación (Ph.D.) degrees from the Polytechnic University of Valencia (UPV), Valencia, Spain, in 1993 and 1998, respectively.

From 1993 to 1995, he was with the R&D Department of the company GH ELIN International (currently GH Group), involved in the design of high current switch-mode power supplies for particle accelerator superconducting magnets. By the end of 1995, he joined the Electronics Engineering Department of the UPV, where he is currently an Associate Professor and the Coordinator of the Industrial Electronic Systems Group (GSEI), <http://gsei.upv.es>. His main research fields are in power converter modeling and control, robust control of switching converters, soft-switching energy conversion, power factor correction circuits, uninterruptible power supplies, grid-connected inverters and converters for renewable energy sources.

Dr. Garcerá is a member of the Spanish Association of Telecommunications Engineers. Since 2004, he has been an Associate Editor of the *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS*.



Emilio Figueres (S'98–A'00–SM'10) received the M.Sc. degree from the Ecole Nationale Supérieure d'Electrotechnique, d'Electronique, d'Informatique et d'Hydraulique de Toulouse, Toulouse, France, in 1995 and the Dr. Ingeniero Industrial (Ph.D.) degree from the Universidad Politécnica de Valencia (UPV), Valencia, Spain, in 2001.

From 1995 to 1996, he was with the R&D Departments of the companies GH Electrotermia (GH Group) and Power Electronics Spain, where he was involved in the design of high-performance power electronic converters. Since 1996, he has been with the Electronics Engineering Department, UPV, where he is currently a Full Professor and the Head of the Department. His main research interests include modeling and control of power converters, power processing of renewable energy sources, and grid-connected converters for distributed power generation and improvement of power quality.



César L. Trujillo Rodríguez received the B.S. degree in electronics engineering from the Universidad Distrital Francisco José de Caldas, Bogotá, Colombia, in 2003, the M.Sc. degree in electrical engineering from the Universidad Nacional de Colombia, Bogotá, in 2006, and he is currently working toward the Ph.D. degree in the Universidad Politécnica de Valencia, Valencia, Spain.

He is an Associate Professor in the Department of Electronic Engineering, Universidad Distrital Francisco José de Caldas, where he currently teaches courses on analog circuits and power electronics. His main research interests include modeling and control of power converters applied to the distributed generation in microgrids.



Javier A. Guacaneme Moreno (S'97–M'11) received the B.S. degree in electronics engineering from the Universidad Distrital Francisco José de Caldas, Bogotá, Colombia, in 1999, the M.Sc. degree in electrical engineering from the Universidad Nacional de Colombia, Bogotá, Colombia, in 2006, and he is currently working toward the Ph.D. degree in the Universidad Politécnica de Valencia, Valencia, Spain.

He is also currently an Associate Professor with responsibility for power electronics, electrical machines, and electrical power systems with the Department of Electronic Engineering at Universidad Distrital Francisco José de Caldas. His research interests are in the areas of high-frequency integrated power conversion applied to renewable energy systems.



David Velasco de la Fuente received the Industrial Engineering degree (M.Sc.) and Project Manager Master degree from the University of Valladolid, Valladolid, Spain, in 2006 and 2007, respectively, and the Electronic System Master degree from the Universidad Politécnica de Valencia, Valencia, Spain, in 2009, where he is currently working toward the Ph.D. degree.

His main research fields are in modeling and control of power converters applied to the distributed generation in microgrids.